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RESEARCH PROPOSAL			

(1) Research Program/Project/Study Title: Influence of golden apple snail (<i>Pomacea canaliculata</i>) as supplement to Darag native chicken (<i>Gallus gallus domesticus</i>)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Dumarao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Agriculture		(8) Discipline/Field of Specialization Animal Science	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Darag chicken is native to the islands of Panay, Guimaras and Negros in the Philippines. Its wild ancestor is the red jungle fowl (*Gallus gallus*) which is pervasive in Southeast Asia. Other similar fowl species that are still in existence are the Ceylones jungle fowl (*Gallus layette*), black or green jungle fowl (*Gallus various*), and the gray jungle fowl (*Gallus sonnerati*). The modern Darag, or the hen, has a yellowish-brown colour. The rooster, which is often called Alimbuyog, has red and black colours. Once it matures, the female Darag can weigh up to an average of one kilogram while males can be as heavy as 1.3 kilograms. According to PSA (2022) the total chicken inventory was registered at 186.36 million birds or a decrease of -0.5 percent compared to stocks in the same period of previous year. Similarly, population of native/improved chicken and layer chicken declined by -1.6 percent and -0.2 percent, respectively.

Pomacea canaliculata, commonly known as the golden apple snail or the channeled apple snail, is a species of large freshwater snail with gills and an operculum, an aquatic gastropod mollusk in the family Ampullariidae, the apple snails. South American in origin, this species is considered to be in the top 100 of the "World's Worst Invasive Alien Species" (Lowe et al. 2000). Golden apple snail (*Pomacea canaliculata*) is a common freshwater snail and a notorious agricultural pest in the Philippines and other countries in Asia (Mochida 1988, 1991; Naylor 1996). However, golden apple snails can also be used as a protein source feed because they have a crude protein (CP) content of more than 20%. The nutrient components of golden apple snail meal is CP 46.2%, metabolism energy (ME) 1920 kcal/kg, calcium (Ca) 2.98% and phosphorus (P) 0.35% (Pena, et. al. 2016). Making silage from apple snail meat has been described as a low cost and simple method for small-scale production (Kaensombath, 2005).

Lactic fermentation is an effeceint method for preserving and improving golden apple snail meat quality. The silage is produced by incubating golden apple snail with locally screened lactic acid bacteria at ambient temperature using molasses as the carbon source for bacterial growth. The results of ensilation of golden apple snail for ten days helped improved the quality of the meal by increasing free amino acids. (Rattanaporn et al. 2006).

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.

This study aims to discover the level of ensiled Golden Apple Snail (*Pomacea canalicuta*) that will significantly enhance the growth performance of Darag Native Chicken. The Golden Apple Snail (*Pomacea canalicuta*) will be cultured to produced homogenous nutrients. Cultured GAS will be collected and washed by potable water and minced to about 1mm to a maximum of 4 mm particle size for the preparation of ensiled golden apple snail. Then, ground GAS will be mixed with sugar cane molasses 18% w/w and lactic acid bacteria 5% w/w. The ensiled GAS will be mixed and then stored in airtight plastic container for 14 days. Rice bran will be added (30%) as a co-drying material for drying the ensiled golden apple snail. Drying of the ensiled golden apple snail will be done in an oven at 60°C until constant weight is reached.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The general objective of the study is to (1) evaluate the growth performance of darag native

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chicken supplemented with different levels of ensiled golden apple snail in terms of: (a) weight gain; (b) feed intake; (c) feed conversion efficiency; (d) dressing percentage; (2) To assess the potential of ensiled golden apple snail as supplement for darag native chicken.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology / information from which the project proposal will take-off).

Native Chicken in the Philippines

The Philippine native chicken is an important livestock resource. They are commonly raised in small groups of up to 24 hens for egg production. Some farmers raise native chickens for meat, barter, or sale as an additional source of income for the family (Santiago, 2018). The native chicken has established a niche market and appears to be unaffected by pressures due to its high quality meat and distinct flavor.

sourced from the commercial layer and broiler markets This makes native chicken production an excellent investment opportunity for small and medium-sized businesses (PCARRD, 2007).

Farming native Darag chicken does not necessitate the same stringent biosecurity and feeding standards that farming commercial varieties, particularly as a contract farmer, does. Thus, Darag native chickens require less upkeep than commercial breeds. They can survive in a natural hardy environment without the need for special housing or feeding. Furthermore, the Darag chickens are disease resistant due to the natural selection process and that they have excellent survivability and can thrive even in the Philippines' harshest environments, where commercial breeds would struggle (Poultry Manual, 2018).

Nutritive Value of Ensiled Golden Apple Snail

The golden apple snail is regarded as a major rice pest in the Philippines and throughout Asia. Its voracious appetite and rapid reproduction rate make this species a nightmare for irrigated rice farmers. There are at least three Pomacea species in the country, which were introduced in various parts of the archipelago in the early 1980s with the goal of developing an escargot industry and increasing farmer income (Adalla & Magsino, 2006).

Budiari et al (2021) reiterates that golden apple snails can be used as a protein source feed since they have a crude protein (CP) content of more than 20%. The nutrient content of golden apple snail meal is 46.2% CP, 1920 kcal/kg ME, 2.98% calcium (Ca), and 0.35% phosphorus (P). Snails had lower methionine amino acid levels than fish meal but higher levels than soybean meal. In broiler chicken, the use of 15% golden snail meal results in a higher feed conversion ratio than the control. The addition of 20% golden snail meal to native chicken rations resulted in the greatest final body weight and the lowest feed conversion.

Roles of Protein to Native Chicken

Protein in the starter phase (0-6 weeks of age) is critical for the survival of native chickens. Amino acids as a constituent of proteins play roles in the growth and physiological functions of chickens. Micronutrients, especially balanced amino acids, can promote maximum growth in native chickens. For economical supplementation to address deficiencies in specific amino acids in poultry feed, synthetic amino acids have been added to chicken feed, particularly beginning in the early 1990s (Vetworld, 2020).

Proteins were required to maintain life carrying cell functions and productivity, such as muscle growth, fat, bone, egg, and seminal fluid/semen (Lesson and Summer, 1991). Protein requirement was equal to amino acid requirement since protein cannot be absorbed directly but have to digest by enzymes become amino acids. Protein requirement was affected by age/ growth phase and feed energy/protein ration. Low protein feed result to protein deficiency on chicken then impact to slow growth, but high protein result to inefficient production. Protein requirement on starter phase will be higher than grower or pullet (Lisnahan, et. al., 2017)

Prior Art Search

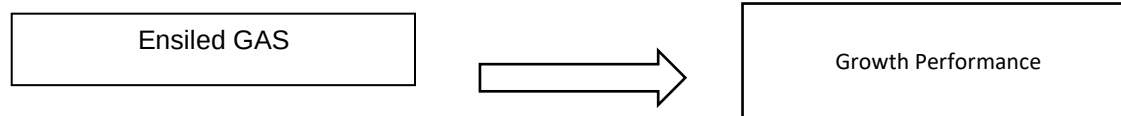
There are already numerous research which are related on the proposed study but based on espacenet and wipo and google scholar, the researcher did not find any research that the ensiled golden apple snail is used in darag native chicken.

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(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

Independent Variable

Dependent Variable



(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Experimental Site

The experiment will be conducted in the Capiz State University from June 2023 to December 2023.

Production of Golden Apple Snail

Three concrete boxes (1mx1mx3m) will be constructed. Water will be provided at 70 cm depth. Water will be change every three days to reduce ammonia build up. In each boxes, 30 golden apple snail will be confined to grow and reproduced. They will be fed with trichantera only every day for 5 months.

Experimental Design and Treatments

One hundred twenty (120) heads of Darag Chicken will be used in this study. The birds will be randomly distributed to assigned cages with recommended space requirement and be given with dietary treatments. Twenty-four (24) cages will be constructed to accommodate one hundred twenty (120) heads of darag chicken. Proper spacing, provision of drinker, and feeding trough will be observed in each cage.

The experiment will consist of four (4) dietary treatments as shown below. It will be laid out in Completely Randomized Design (CRD). This will be replicated six (6) times with five (5) birds per cage.

- | | |
|---|---|
| A | 100% commercial feeds |
| B | 95% commercial feeds + ensiled GAS at 5% inclusion |
| C | 90% commercial feeds + ensiled GAS at 10% inclusion |
| D | 85% commercial feeds + ensiled GAS at 15% inclusion |

Making ensiled kuhol

Step 1. Collect the kuhol and wash with potable water

Step 2. Dip into boiling water for 2 min. and removed the meat from the shell

Step 3. Cut and mince the kuhol about 2mm in size.

Step 4. Weigh the minced kuhol

Step 5. Add 18% of molasses or brown sugar w/w , then 5% of lactic acid bacteria w/w, and 30% of rice bran w/w, and mixed together in a bowl or large pan. Coat as much of the surface area of the kuhol with molasses asmuch as possible.

Step 6. Pack the mixture into an air-tight container and wait for 14 days

Step 7. After 14 days placed the ensiled GAS into oven then set to 60° C until the constant weight is reached.

Step 8. When the constant weight is reach, transfer the dry silage into mortar and pestle and manually grind your GAS dried silage.

Step 9. Store the GAS powder properly. Transfer the ensiled GAS powder into a glass or food-grade polyethylene (PE) plastic container.

[illegible]

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Sub-total		270,462.00
B. Personal Services		
Salaries		
RA (11,068/mo x 6 mos.)	1	66,408.00
Laborer (6,000/mo x 24 mos.)		
Honoraria		
Study Leaders (8,000/mo x 24 mos.)		
Sub-total		66,408.00
C. MOOE		
Travel/Transportation Expenses		10,000.00
Housing and Facilities		300,000.00
Communication Expenses		3,000.00
Postage and deliveries, Telephone Expenses, Internet Expenses, etc.		
Supplies and Materials Expenses		
Office Supplies		5,000.00
Representation Expenses (food for meetings, etc.)		
Printing and Binding Expenses		5,000.00
Sub-total		223,000.00
Total		659,870.00
(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)		
Adalla, C.A.; Magsino, E.A., Understanding the golden apple snail (<i>Pomacea canaliculata</i>): biology and early initiatives to control the pest in the Philippines, 2016, Philippines Univ. Los Banos, College, Laguna (Philippines). Crop Protection Cluster; Philippine Rice Research Inst., Science City of Munoz,		

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Nueva Ecija (Philippines)

Kaensombath, L., 2005. Evaluation of the nutritive value of ensiled and fresh Golden Apple snails (*Pomacea* spp) for growing pigs. MSc. Thesis, Swedish University of Agricultural Science

Steven Leeson, Ph.D., and John D. Summers. 2008, Commercial Poultry Nutrition Third Edition, ERSITY Books P. O. Box 1326 Guelph, Ontario, Canada N1h 6n8 Nottingham University Press Manor Farm, Church Lane, Thrumpton, Nottingham, Ng11 0ax, England

Lisnahan, C.V., Wihandoyo, Zuprizal, Harimurti, S., 2017, Effect of addition of methionine and lysine into diets based on cafeteria standards on the growth performance of native chickens at starter phase, International Journal of Poultry Science

Lowe, S., et al. (2000). 100 of the world's worst invasive alien species: a selection from the global invasive species database. Auckland: Invasive Species Specialist Group Species Survival Commission, World Conservation Union (IUCN); 2000.

MOCHIDA O. 1988. Nonseedborne rice pests of quarantine importance. In: Rice Seed Health. Los Baños, Philippines: International Rice Research Institute. p. 117-129.

MOCHIDA O. 1991. Spread of freshwater *Pomacea* snails (Pilidae, Mollusca) from Argentina to Asia. Micronesia Suppl 3:51-62

NAYLOR R. 1996. Invasion in agriculture: Assessing the cost of the golden apple snail in Asia. Ambio 25: 443-448.

Philippine Council for Agriculture, Forestry and Natural Resources Research and Development (PCARRD), 2007, Profitability Analysis: 200-hen Module Native Chicken Production, Department of Science and Technology (DOST), Los Baños, Laguna

Philippine Statistic Authority., 2007, Chicken situation report April To June 2022.

Poultry manual, 2018 <https://poultrymanual.com/rearing-daraq-native-chickens-in-the-philippines>

Rattanapon, et al.. (2006). Effect of molasses on golden apple snail silage production. In: processings of the 44 Kasetsart University Annual Conference, Bangkok, Thailand, pp. 3-10.

[Vet World](#). 2020 Dec; 13(12): 2765–2771.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7811536/#:~:text=Protein%20in%20the%20starter%20phase,maximum%20growth%20in%20native%20chickens.>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

REYNALD A. TOMOGDA

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EDUCATION

GRADUATE STUDIES: MS in Animal Science

Capiz State University, Burias Campus Currently studying

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COLLEGE: Bachelor of Science in Agriculture
Capiz State University, Pontevedra Campus AY 2018

HIGH SCHOOL: St. Joseph Diocesan School
Cabug-cabug, President Roxas, Capiz AY 2014

ELEMENTARY: Cubay-Ibaca Elementary School AY 2010

WORK EXPERIENCE:

Baranggay Recor Keeper @ Cubay Baranggay Government Unit from July 2018- October 2018
Livestock Technician @ Office of the Provincial Veterinarian, Capiz Provincial Government from November 08, 2018- August 12, 2022
Faculty @ Capiz State University, Dumarao Satellite College from August 15, 2022 to present